QML Lab • research project 15

Barren-Plateau Mitigation

Comparing strategies that keep gradients alive as circuits grow.

Living document • last updated October 25, 2026

Quadrant	CQ (classical data, quantum processor)	**Track**	Noise and hardware
Difficulty	Medium	**Tools**	PennyLane
Builds on	Module 8 of the QML Lab	**Workspace**	`qml_projects/barren-plateau-mitigation/`

Summary. Module 8 showed that gradients of random deep circuits vanish exponentially with the number of qubits. This project compares the main remedies, local cost functions, small-angle and identity-block initialisation, and layerwise training, by gradient variance at initialisation and by what they achieve after a fixed training budget. It also asks the uncomfortable question raised by recent theory: are the circuits that avoid barren plateaus the ones a classical computer can simulate?

1 Background and key concepts

1.1 Barren plateaus

For a cost $C(\theta) = \mathrm{Tr}[O\,U(\theta)\rho U(\theta)^\dagger]$ and sufficiently random circuits, McClean et al. (2018) showed

$$\mathbb{E}_\theta[\partial_k C] = 0, \qquad \mathrm{Var}_\theta[\partial_k C] \in O(b^{-n}),\ b > 1. \tag{1}$$

Exponentially many shots would be needed to estimate a gradient. Cerezo et al. (2021) showed that the choice of cost matters: global observables (acting on all qubits) give barren plateaus even for shallow circuits, local observables on shallow circuits keep gradients polynomially large. Holmes et al. (2022) linked the effect to expressibility: the closer the circuit ensemble is to random unitaries, the flatter the landscape. Noise adds its own plateau (Wang et al., 2021).

1.2 Mitigation strategies

Identity-block initialisation
Grant et al. (2019) build the circuit from blocks $U_b(\theta_b)\,U_b(\theta'_b)^\dagger$ initialised with $\theta'_b = \theta_b$, so the whole circuit starts as the identity and early gradients are large.

Small-angle initialisation
Drawing $\theta \sim \mathcal{N}(0, \sigma^2)$ with small σ keeps the circuit near the identity at the start.

Layerwise learning
Skolik et al. (2021) train a few layers at a time and freeze them, adding depth gradually.

Local costs
Replace a global observable such as $Z^{\otimes n}$ by a sum of local ones.

1.3 The simulability caveat

Cerezo et al. (2023) argue that many architectures with provable absence of barren plateaus can be simulated efficiently classically, because their relevant dynamics lives in a small subspace. A complete study should therefore ask, for each successful strategy, whether a classical surrogate could reproduce it.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Which strategy keeps gradient variance flattest as the number of qubits grows from 2 to 12?

RQ2 Does a larger initial gradient translate into a lower final loss after a fixed training budget?

RQ3 How do strategies combine, for example local cost with identity-block initialisation?

3 Method

3.1 Data

Two training tasks: learning to return a randomly prepared state to $|0 \cdots 0\rangle$ (a controllable state-learning problem), and the Module 5 classifier on two moons with increasing qubit numbers.

3.2 Models

The hardware-efficient ansatz of Module 8 with $L = n$ layers, extended with the parameters `init_strategy` (uniform, small-angle, identity-block), `cost` (global, local) and `training` (all layers, layerwise). Gradient variances from 200 random initialisations per setting.

3.3 Baselines

Module 8's configuration (uniform initialisation, global cost) as the reference; analytic $O(2^{-n})$ scaling as a guide.

3.4 Metrics

- $\mathrm{Var}[\partial_k C]$ versus n on a log scale, with fitted decay rate.
- Final loss and steps to reach a threshold after a fixed budget.
- Evidence for or against classical simulability of the successful settings.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Reproduce Module 8	$n = 2..12$, uniform, global	variance slope
E2	Initialisation	uniform, small-angle, identity-block	variance, final loss
E3	Cost locality	global vs local	variance, final loss
E4	Layerwise learning	layers added per stage	final loss
E5	Combinations and noise	best pairs, depolarizing p	final loss

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/barren-plateau-mitigation/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/barren-plateau-mitigation/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "init-strategies"
title = "Gradient variance by initialisation strategy"
script = "experiments/e01_init_strategies.py"
figure = "variance_by_strategy.png"

[experiments.params]
max_qubits = 12
init_strategy = "identity_block"
cost = "local"
samples = 200
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/barren-plateau-mitigation
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_init_strategies.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Report both gradient variance at initialisation and final loss; one without the other is misleading.

7 Open directions

- Symmetry-respecting (equivariant) ansätze.
- Warm starts from classical solutions or from smaller circuits.
- Classical surrogates for the trainable regimes, testing the simulability argument directly.

8 Risks and limitations

- Large initial gradients do not guarantee good final solutions.
- Identity-block initialisation helps only at the start of training.
- State-vector memory limits the study to about 14 qubits.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

Cerezo, M., Sone, A., Volkoff, T., Cincio, L. and Coles, P. J. (2021) 'Cost function dependent barren plateaus in shallow parametrized quantum circuits', *Nature Communications*, 12, 1791. Available at: https://doi.org/10.1038/s41467-021-21728-w.

Cerezo, M., Larocca, M., García-Martín, D., Diaz, N. L., Braccia, P. et al. (2023) 'Does provable absence of barren plateaus imply classical simulability? Or, why we need to rethink variational quantum computing', Preprint, arXiv:2312.09121. Available at: https://arxiv.org/abs/2312.09121.

Grant, E., Wossnig, L., Ostaszewski, M. and Benedetti, M. (2019) 'An initialization strategy for addressing barren plateaus in parametrized quantum circuits', *Quantum*, 3, 214. Available at: https://doi.org/10.22331/q-2019-12-09-214.

Holmes, Z., Sharma, K., Cerezo, M. and Coles, P. J. (2022) 'Connecting ansatz expressibility to gradient magnitudes and barren plateaus', *PRX Quantum*, 3(1), 010313. Available at: https://doi.org/10.1103/PRXQuantum.3.010313.

McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R. and Neven, H. (2018) 'Barren plateaus in quantum neural network training landscapes', *Nature Communications*, 9, 4812. Available at: https://doi.org/10.1038/s41467-018-07090-4.

Skolik, A., McClean, J. R., Mohseni, M., van der Smagt, P. and Leib, M. (2021) 'Layerwise learning for quantum neural networks', *Quantum Machine Intelligence*, 3, 5. Available at: https://doi.org/10.1007/s42484-020-00036-4.

Wang, S., Fontana, E., Cerezo, M., Sharma, K., Sone, A., Cincio, L. and Coles, P. J. (2021) 'Noise-induced barren plateaus in variational quantum algorithms', *Nature Communications*, 12, 6961. Available at: https://doi.org/10.1038/s41467-021-27045-6.